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Analysing the Generalisation and Reliability of Steering Vectors

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Abstract

Steering vectors (SVs) are a new approach to efficiently adjust language model behaviour at inference time by intervening on intermediate model activations. They have shown promise in terms of improving both capabilities and model alignment. However, the reliability and generalisation properties of this approach are unknown. In this work, we rigorously investigate these properties, and show that steering vectors have substantial limitations both in- and out-of-distribution. In-distribution, steerability is highly variable across different inputs. Depending on the concept, spurious biases can substantially contribute to how effective steering is for each input, presenting a challenge for the widespread use of steering vectors. We additionally show steerability is also mostly a property of the dataset rather than the model by measuring steerability across multiple models. Out-of-distribution, while steering vectors often generalise well, for several concepts they are brittle to reasonable changes in the prompt, resulting in them failing to generalise well. Similarity in behaviour between distributions somewhat predicts generalisation performance, but there is more work needed to understand when and why steering vectors generalise correctly. Overall, our findings show that while steering can work well in the right circumstances, there remain many technical difficulties of applying steering vectors to robustly guide models' behaviour at scale.

1 Introduction

Steering Vectors (SVs) [30, 33, 43, 18] have been recently proposed as a technique for guiding language model behaviour at inference time. Existing work has shown promising results in using these SVs to detect and guide models towards high-level traits such as honesty [43], sycophancy [30], and positive sentiment [31]. They have also been shown to be useful for improving model capabilities [39, 18, 34] and red-teaming [29]. SVs are of interest as they enjoy a number of practical benefits over other model adjustment techniques that require adding more information into the context window [5, 38, In-Context Learning], or performing training to adjust model parameters (fine-tuning). Recent work shows that steering vectors can be learned in an unsupervised way [20], thus removing another obstacle for their use. It may even be possible for different steering vectors to be used in

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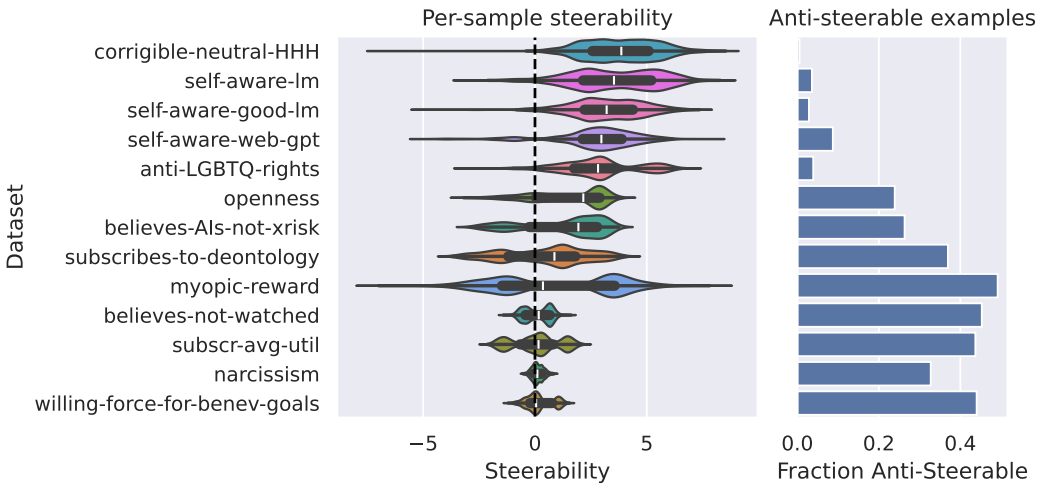

Figure 1: **Steering effects are not reliable, and often steer in the opposite direction**. We show per-sample steerability and the fraction of anti-steerable examples for a representative sample of 13 datasets (out of 40 total). Many dataset have a high variation in per-sample steerability, and several datasets produce the opposite behaviour for almost 50% of inputs. For all datasets see Figure 18. Some dataset names have been shortened.

combination for multiple behaviours [34, 36]. It would thus be very important and useful in practice if steering vectors were truly effective.

However, existing work has mostly evaluated SVs in-distribution, and looked at aggregate behaviour. It is unknown how reliable the change in behaviour caused by SVs is, and how well SVs generalise to different system or user prompts. In this paper, we extensively evaluate the in-distribution reliability and out-of-distribution generalisation of SVs, extending analysis in Rimsky et al. [30] to a much broader variety of behaviours from the Model Written Evals (MWE) datasets by Perez et al. [26]. In addition, we consider targeted distribution shifts in the form of inserting prompts via the user message or system message. This setting mimics the practically important setting where we will need to apply SVs to different system and user prompts, and where we would require SVs to generalise well to be robustly useful.

Our first key result is that **for many behaviours studied, steering is unreliable** (Figure 1 and Section 5). For all behaviours evaluated, steerability takes on a large range of values across different inputs, including negative values, where SVs produce the opposite of the desired behaviour. Previous work [30, 33, 43, 18] does not study this variance, which potentially leads to over-optimistic claims on performance due to a lack of error bars. In explaining this variance, we demonstrate a novel type of bias, *steerability bias*, in which models are easier to steer towards outputs with a certain property (i.e. answer position or token choice). The lack of steerability and high variance in steering performance demonstrates that in many cases, a steering vector extracted may not correspond to the intended concept, and applying steering vectors may only be effective in the presence of spurious factors associated with the prompt template or other potential biases.

Our second set of results focuses on the out-of-distribution setting. Here, we find that **SVs generalise reasonably well across different prompt settings, but the generalisation behaviour is not perfect or entirely predictable** (Section 6). SVs generalise better over some shifts than others and generally perform worse out-of-distribution vs in-distribution. We investigate what causes this difference in generalisation properties, finding that (i) steerability is mostly a dataset-level property, with similar datasets being steerable and producing generalisable SVs for two different models; and (ii) SVs generalise better when model behaviour is similar in the source and target prompt setting. This relationship is a potential issue for SVs, as SVs will need to be applied to guide models towards behaviours they do not normally produce.

Overall, our findings indicate that steering vectors in their current form are not a panacea for aligning model behavior at inference time. Despite their promise, more work is required to ensure that steering vectors reliably produce the desired behaviour in a generalisable way and are practically useful.

2 Related Work

Steering Vectors (SVs, also known as activation engineering) and related ideas were introduced by Turner et al. [33], Zou et al. [43], Liu et al. [18]. SVs can be seen as an inference-time intervention [16] technique in the representation engineering [43] toolkit, which is an umbrella term for the broad approach of improving the transparency and controllability of neural networks by examining and intervening on population-level representations and activations of the network. [28]. Rimsky et al. [30] recently introduced Contrastive Activation Addition (CAA), a specific technique for extracting and applying SVs which we use in this work, due to its simplicity, effectiveness and popularity in the community. Rimsky et al. [30] demonstrate the effectiveness of CAA in-distribution on several AI alignment-relevant behaviours, while we test on a much broader range of behaviours, investigate the reliability of the steering intervention, and examine out-of-distribution generalisation of SVs. We describe the CAA method in more detail in Section 3.

Compared to fine-tuning [42, 27, 23], steering vectors don't involve changing model parameters, hence potentially avoiding catastrophic forgetting [3, 19]. Compared to in-context learning [5, 38, ICL], steering does not require adding tokens to the prompt, saving inference cost and enabling it to scale beyond the length of the context window. Furthermore, Zou et al. [43] show that steering interventions are robust to adversarial attacks capable of breaking prompt-based and fine-tuning-based alignment methods [26, 37, 13]. An extended related work section including discussing the relationship between SVs and the Linear Representation Hypothesis [25, LRH] and other works that evaluate the generalisation behaviour of model adjustment methods can be found in Appendix B.

3 Preliminaries

Rimsky et al. [30] propose Contrastive Activation Addition (CAA) to extract and apply steering vectors on datasets. We follow this protocol in our experiments, and so we summarise the main steps here.

Multiple-Choice Contrastive Prompts. We construct a prompt consisting of a question or statement followed by two multiple-choice options labelled "(A)" and "(B)". The model is tasked with reading the question and available options (x), then choosing one of the options (y_+ or y_-). For some datasets these two options are statements, and for others the two options are either "Yes" or "No". A typical example is shown in Figure 9. During preprocessing, we randomise whether 'A' or 'B' (and 'Yes' or 'No' where appropriate) are used as the positive y_+ or negative y_- options, to ensure that we do not simply extract a steering vector for e.g. the token 'A' vs the token 'B'.

Steering Vector Extraction. For a given dataset $\mathcal{D}$ consisting of triples of the form (x, y_+, y_-), and a given layer L, activations are extracted from the residual stream at the multiple-choice option token position for the positive and negative option, to get $a_L(x, y_+)$ and $a_L(x, y_-)$ respectively. We extract a steering vector v_{MD} using the mean difference (MD) of positive and negative activations:

$$v_{MD} = \frac{1}{|\mathcal{D}|} \sum_{(x, y_-, y_+) \in \mathcal{D}} \left[a_L(x, y_+) - a_L(x, y_-) \right] \tag{1}$$

We note that other aggregation methods have been proposed, but literature does not suggest these perform better than mean-difference. We discuss alternatives in Appendix D.4.

Steering Intervention. To apply a steering intervention at layer L using a steering vector v_L, we add $\lambda * v_L$ into the activations at the last token position at layer L during model inference. Here, λ is a multiplier that controls the strength of the steering intervention. For any metric of (change in) behaviour, we can evaluate that metric for a range of λs to ascertain the effectiveness of a steering intervention; more details including our specific choice of metric are discussed subsequently in Section 4.2